

Employee Salary Management System

Experiment No: 01

Date: 11/09/2026

Title: Employee Salary Management System using Interfaces

Objective / Aim

To design and implement an Employee Salary Management System in Java using interfaces and classes, demonstrating interface implementation, method overriding, and runtime polymorphism to calculate salary details for regular and contract employees.

System Requirements

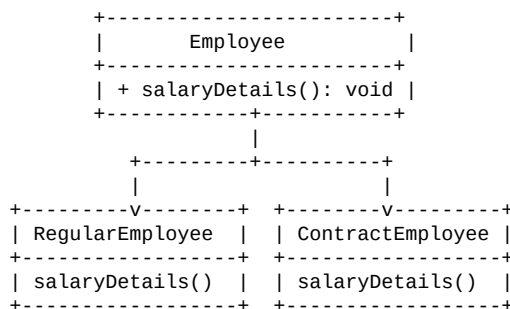
Hardware: Standard PC/Laptop; minimum 4 GB RAM.

Software: Windows / Linux / macOS; JDK 17 or JDK 21; IDE such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code.

Algorithm / Logic Description

1. Create an interface named Employee.
2. Declare a method salaryDetails() in the interface.
3. Create RegularEmployee implementing Employee.
4. Calculate regular salary using basic pay, HRA, and TA.
5. Create ContractEmployee implementing Employee.
6. Calculate contract salary using basic pay, HRA, and TA.
7. Accept the employee ID from the user.
8. If the ID starts with R, create a RegularEmployee object.
9. If the ID starts with C, create a ContractEmployee object.
10. Display salary details or an invalid employee ID message.

UML Class Diagram



Relationship: RegularEmployee and ContractEmployee implement the Employee interface.

Source Code

```
package javacore;

import java.util.Scanner;

interface Employee {
    void salaryDetails();
}

class RegularEmployee implements Employee {
    public void salaryDetails() {
        int basic = 25000, hra = 15000, ta = 5000;
        int total = basic + hra + ta;

        System.out.println("Salary Details:");
        System.out.println("Basic Pay: " + basic);
        System.out.println("HRA: " + hra);
    }
}
```

```

        System.out.println("T.A: " + ta);
        System.out.println("Total Amount: " + total);
    }
}

class ContractEmployee implements Employee {
    public void salaryDetails() {
        int basic = 12000, hra = 0, ta = 3000;
        int total = basic + hra + ta;

        System.out.println("Salary Details:");
        System.out.println("Basic Pay: " + basic);
        System.out.println("HRA: " + hra);
        System.out.println("T.A: " + ta);
        System.out.println("Total Amount: " + total);
    }
}

public class EmployeeSalary {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Employee Id: ");
        String id = sc.nextLine();

        Employee e;

        if (id.startsWith("R")) {
            e = new RegularEmployee();
            e.salaryDetails();
        } else if (id.startsWith("C")) {
            e = new ContractEmployee();
            e.salaryDetails();
        } else {
            System.out.println("Invalid Employee Id");
        }

        sc.close();
    }
}

```

Compilation & Execution Commands

Save as: EmployeeSalary.java

```
javac EmployeeSalary.java
java javacore.EmployeeSalary
```

Output Screen

Case 1: Regular Employee

```
Enter Employee Id: R101
Salary Details:
Basic Pay: 25000
HRA: 15000
T.A: 5000
Total Amount: 45000
```

Case 2: Contract Employee

```
Enter Employee Id: C101
Salary Details:
Basic Pay: 12000
HRA: 0
T.A: 3000
Total Amount: 15000
```

Case 3: Invalid Employee ID

```
Enter Employee Id: X101
Invalid Employee Id
```

Result

The Employee Salary Management System was successfully implemented in Java using an interface and classes. The program displays salary details for regular and contract employees based on the employee ID entered by the user.